Determine whether the following assertions are sometimes, always, or never true. Justify your answers. This means giving a proof is your answer is "always" or "never", and giving an example of the statement being true and an example of it being false if your answer is "sometimes."

1. The sum of two rational numbers is rational.

2. The product of two rational numbers is rational.

3. The sum of a rational number and an irrational number is irrational.

4. The product of a rational number and an irrational number is irrational.

5. The product of two irrational numbers is irrational.

6. The sum of two irrational numbers is irrational.